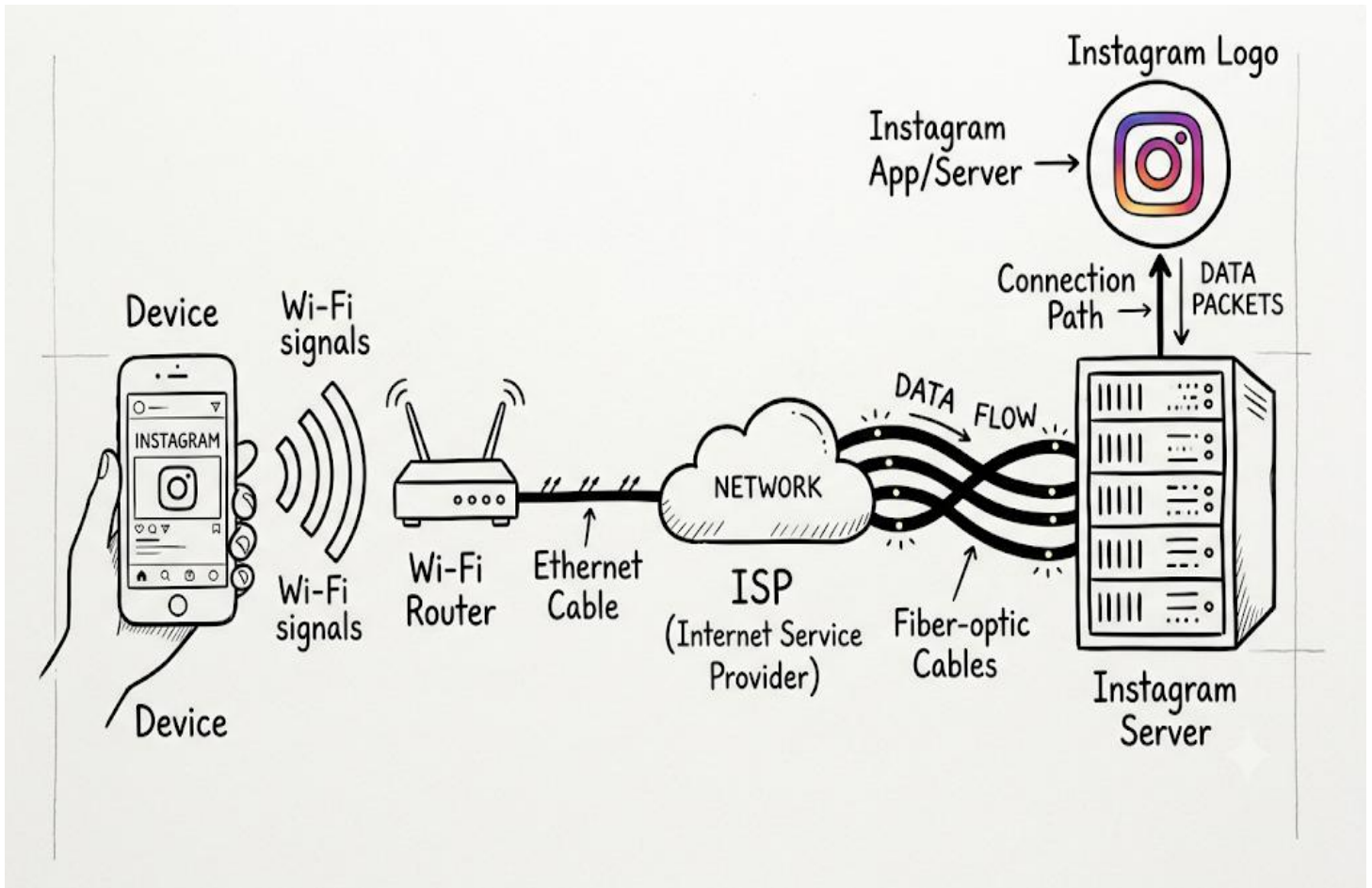


INTRODUCTION TO NETWORKS

1. DIAGRAM OF CONNECTION :



2. DAILY NETWORK DEVICE AND NETWORK TYPE :

DEVICE NAME	NETWORK TYPE CONNECTED TO	PRIMARY PURPOSE / FUNCTION
SMARTPHONE	WAN (VIA 4G/5G CELLULAR) & LAN (VIA Wi-Fi)	ACCESSES EXTERNAL INTERNET RESOURCES AND COMMUNICATES WITH LOCAL NETWORK DEVICES.
SMARTWATCH	PAN (PERSONAL AREA NETWORK VIA BLUETOOTH)	PAIRS DIRECTLY WITH A SMARTPHONE TO SYNC NOTIFICATION AND HEALTH TRACKING DATA SHORT-RANGE.
LAPTOP	LAN (LOCAL AREA NETWORK VIA Wi-Fi/ETHERNET)	CONNECTS TO HOME/OFFICE LOCAL NETWORKS TO SHARE PRINTERS, FILES, AND ROUTE INTERNET TRAFFIC.
Wi-Fi ROUTER	LAN (INTERNAL FACING) & WAN (EXTERNAL FACING)	ACTS AS THE BRIDGE THAT LINKS THE LOCAL AREA NETWORK (HOME) TO THE WIDE AREA NETWORK (ISP).
SMART TV	LAN (LOCAL AREA NETWORK)	STREAMS VIDEO CONTENT OVER LOCAL Wi-Fi AND ALLOWS MEDIA CASTING FROM OTHER LOCAL DEVICES.

3. DIFFERENCE BETWEEN THE INTERNET AND AN INTRANET :

- **DEFINITIONS**
- **INTERNET:** A GLOBALLY CONNECTED PUBLIC NETWORK OF COMPUTERS ACCESSIBLE TO ANYONE WORLDWIDE.
- **INTRANET:** A PRIVATE, RESTRICTED NETWORK ACCESSIBLE ONLY TO AUTHORIZED INTERNAL USERS WITHIN A SPECIFIC ORGANIZATION.

- **COMPARISON TABLE**

FEATURE	THE INTERNET	INTRANET
ACCESSIBILITY	PUBLIC (ANYONE WITH A CONNECTION)	PRIVATE (AUTHORIZED EMPLOYEES/STUDENTS ONLY)
SECURITY	LOWER (REQUIRES FIREWALLS AND ENCRYPTION)	HIGH (ENCLOSED WITHIN PRIVATE FIREWALLS)
COVERAGE AREA	WORLDWIDE (GLOBAL WAN)	LOCALIZED (SINGLE CAMPUS, OFFICE, OR BUILDING)

4 DATA FLOW :

- **DEVICE (YOUR PHONE):** YOU OPEN THE INSTAGRAM APP AND TAP TO LOAD A PHOTO OR VIDEO.
- **Wi-Fi ROUTER:** YOUR PHONE SENDS THE REQUEST THROUGH INVISIBLE WIRELESS SIGNALS (Wi-Fi) TO YOUR HOME ROUTER.
- **ISP (INTERNET SERVICE PROVIDER):** THE ROUTER SENDS THE REQUEST THROUGH WIRED CABLES TO YOUR INTERNET PROVIDER (LIKE JIO OR AIRTEL), WHICH DIRECTS IT ACROSS THE INTERNET.

- **INSTAGRAM SERVER:** THE REQUEST REACHES INSTAGRAM'S COMPUTER, WHICH FINDS YOUR FEED AND SENDS THE DATA BACK ALONG THE SAME PATH TO YOUR SCREEN.
- **SECURITY & ENCRYPTION:** BEFORE DATA LEAVES YOUR PHONE, IT IS SCRAMBLED (ENCRYPTED) INTO SECURE CODE SO HACKERS OR OUTSIDE NETWORKS CANNOT READ YOUR PRIVATE MESSAGES OR LOGIN DETAILS.
- **DATA PACKETS:** LARGE IMAGES AND VIDEOS ARE BROKEN DOWN INTO THOUSANDS OF TINY PIECES CALLED "DATA PACKETS" THAT TRAVEL INDIVIDUALLY ACROSS THE NETWORK AND REASSEMBLE ON YOUR SCREEN IN MILLISECONDS.